

Hardware IP Card

Basic Information		
Licensing		Open-source (Apache-2.0)
Target FPGA or ASIC		FPGA, ASIC
Design Language		SystemC, SystemVerilog
System-Level Features		
Coupling Type		Loosely-coupled
Data Types		uint64
ISA		RISC-V
Peak Ops Per Cycle		1 event/cycle
Interfaces		- TIP (Trace Ingress Port) (Observer, optional) - ITI (Instruction Trace Interface) (Observer, optional) - Camera (Observer, optional) - ROS (Observer, optional) - Platon_HW (Observer, optional)
Supported Operations		Timing contract monitoring (Periodic, Reaction, Aging, TSBC, TSBC PC)
Compiler Support		TCCP-CO Compiler
External Memory Required		false
Cache Or Scratchpad		DPRAM
Architecture		
Type		Coprocessor / Monitoring Unit
Num Clock Domains		1+, tested with 6
Control Interface	ISAExtension	
Control Interface	Memory Mapped	false
Control Interface	Protocol	DPRAM interface for configuration loading
Memory Interface	Interface Protocol	DPRAM
Memory Interface	Hierarchy Level	
Memory Interface	Cached	false
Memory Interface	IOMMU	false
Interrupts		Yes (violation and fifo_overflow signals)
Microarchitecture		
Parametrization	Parametric Num Units	Yes (configurable TRACEIFs and CONTRACTS template parameters)
Parametrization	Parametric Config	Yes (number of trace interfaces and watchdog contracts)
Programmability	Programmable Cores	false
Programmability	ISAExtension	
Software		
Compiler	Requires Specialized Compiler	true
Compiler	Supported Compilers	TCCP-CO

Hardware Abstraction Layer	HAL Functions	
highLevelAPI	Has High Level API	false
highLevelAPI	SDK Repository	
highLevelAPI	Has Domain Specific Compiler	true
Integration		
IP Distribution	Manifest Type	Makefile
IP Distribution	Standalone Simulation	true
IP Distribution	Standalone Simulation SW Requirements	SystemC 2.3.3, Make, GCC/Clang
IP Distribution	Integration Docs Or Examples	README.md, SystemC/TCCP/README.md
Synthesis	Is Synthesizable	true
Synthesis	FPGA Synthesis Examples	Only as prototype
Synthesis	ASIC Synthesis Examples	
Simulation	Closed Source Simulation	
Simulation	Open Source Simulation	SystemC simulation, SystemVerilog simulation only as prototype
Evaluation	PPA Results Available	true
Physical Implementation		
ASIC Area	Num Gates	
ASIC Area	On Chip Internal Memory Size	
ASIC Area	Target Clock Frequency	